

GRADE 10 • ALGEBRA II • TOPIC 2

Lessons 2-1 and 2-2

Vertex Form and Standard Form

Quadratic Functions and Equations

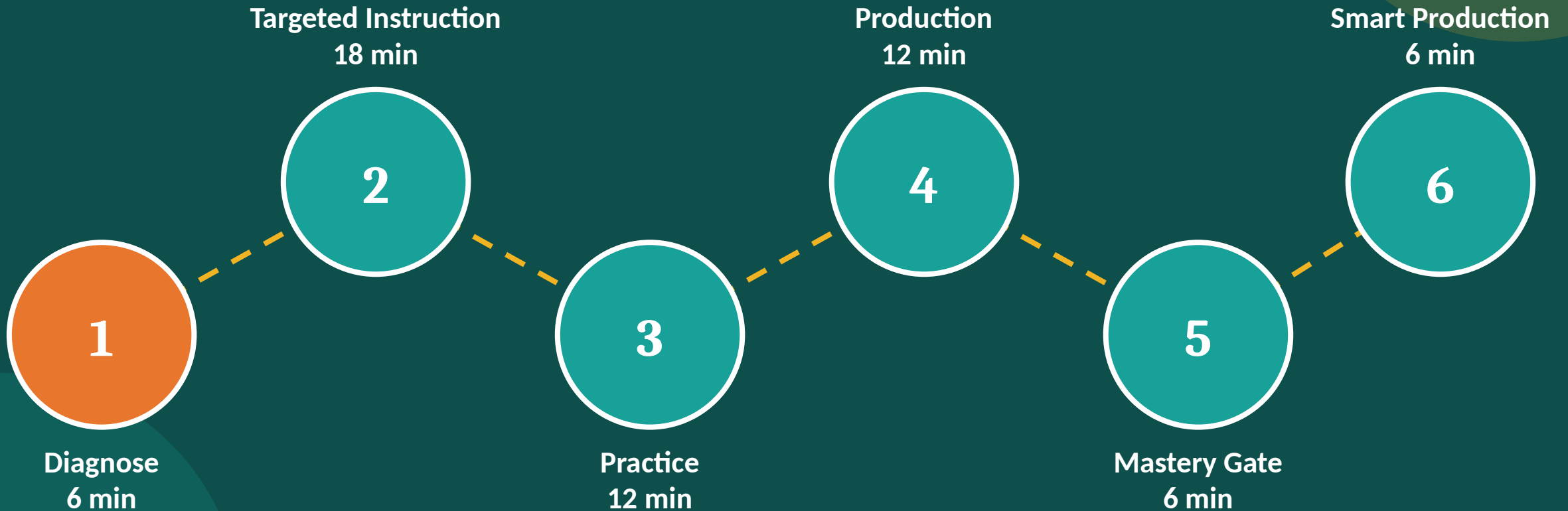
Mr Malek Thiab

$$a(x - h)^2 + k = ax^2 + bx + c$$

Journey of 6 stations, 60 minutes

Today's journey

Six stations. Each one zooms in as we arrive.



What we are learning

OBJECTIVES

1. Graph quadratic functions given in vertex form.
2. Write a quadratic function in vertex form given its key features.
3. Write a quadratic function in standard form and analyze the significance of each term.
4. Identify and explain the key features of quadratic functions in standard form.
5. Graph quadratic functions using key features and symmetry to accurately represent their shape and position on the coordinate plane.

VOCABULARY

axis of symmetry

maximum point

maximum value

minimum point

minimum value

Parabola

parent quadratic function

quadratic function

Vertex

vertex form

constant term

leading coefficient

leading term

linear coefficient

linear term

standard form (of a quadratic function)

Standards

HSA.REI.B.4 Solve quadratic equations in one variable.

HSA.REI.B.4.A Use the method of completing the square to transform any quadratic equation in x into an equation of the form $(x - p)^2 = q$ that has the same solutions. Derive the quadratic formula from this form.

HSF.IF.C.7.A Graph linear and quadratic functions and show intercepts, maxima, and minima.

HSA.CED.A.2 Create equations in two or more variables to represent relationships between quantities; graph equations on coordinate axes with labels and scales.

HSF.IF.B.4 For a function that models a relationship between two quantities, interpret key features of graphs and tables in terms of the quantities, and sketch graphs showing key features given a verbal description of the relationship. Key features include: intercepts; intervals where the function is increasing, decreasing, positive, or negative; relative maximums and minimums; symmetries; end behavior; and periodicity.

HSF.BF.B.3 Identify the effect on the graph of replacing $f(x)$ by $f(x) + k$, $k f(x)$, $f(kx)$, and $f(x + k)$ for specific values of k (both positive and negative); find the value of k given the graphs. Experiment with cases and illustrate an explanation of the effects on the graph using technology. Include recognizing even and odd functions from their graphs and algebraic expressions for them.

HSF.IF.C.7 Graph functions expressed symbolically and show key features of the graph, by hand in simple cases and using technology for more complicated cases.

HSS.ID.B.6.A Fit a function to the data; use functions fitted to data to solve problems in the context of the data. Use given functions or choose a function suggested by the context. Emphasize linear, quadratic, and exponential models.

STATION 1 • DIAGNOSE • 6 MIN

Start here: three quick checks

1

$$(x - 3)^2 = ?$$

2

$$f(x) = x^2 - 6x + 5 : f(0) = ?$$

3

$$y = (x + 2)^2 - 5 \text{ is } y = x^2 \text{ moved ?}$$

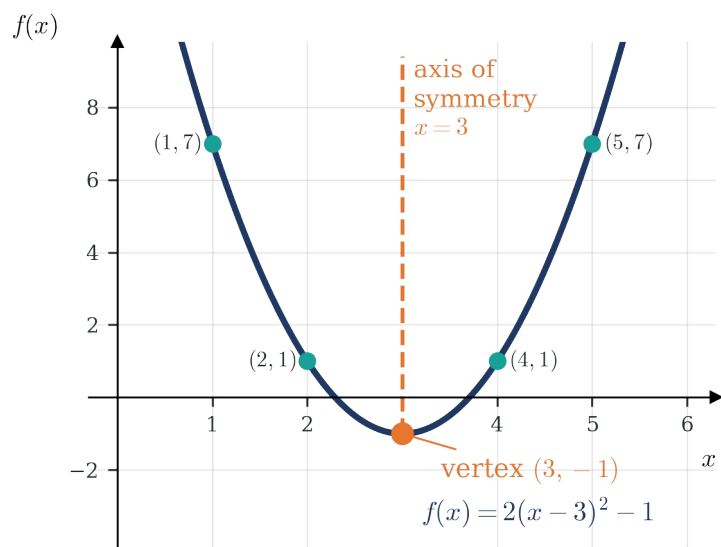
Done when

All three answers are written without a calculator, and you can say what $f(0)$ tells you about the graph.

STATION 2 • TARGETED INSTRUCTION • 18 MIN

Vertex Form: Read the Turning Point

$$f(x) = a(x - h)^2 + k, \quad \text{vertex } (h, k), \quad x = h \quad a > 0: \text{ opens up (min } k) \quad a < 0: \text{ opens down (max } k)$$



EXAMPLE A

Graph $f(x) = 2(x - 3)^2 - 1$

Vertex $(3, -1)$, axis $x = 3$

$$f(2) = f(4) = 2(1)^2 - 1 = 1$$

$$f(1) = f(5) = 2(4) - 1 = 7$$

EXAMPLE B

$f(x) = -(x + 2)^2 + 5$:
vertex, direction, extreme value
 $h = -2, k = 5 \Rightarrow (-2, 5)$
 $a = -1 < 0$: opens down
maximum value 5 at $x = -2$

STATION 2 • TARGETED INSTRUCTION • 18 MIN

Writing Vertex Form from Key Features

$$f(x) = a(x - h)^2 + k \quad 1. \text{ Put the vertex } (h, k) \text{ in. } 2. \text{ Use the extra point to find } a$$

EXAMPLE A

Vertex $(-2, 5)$, through $(0, -3)$

$$f(x) = a(x + 2)^2 + 5$$

$$-3 = a(2)^2 + 5 \Rightarrow a = -2$$

$$f(x) = -2(x + 2)^2 + 5$$

EXAMPLE B

Maximum value 12 at
 $x = 3$, through $(1, 4)$

$$f(x) = a(x - 3)^2 + 12$$

$$4 = a(-2)^2 + 12 \Rightarrow a = -2$$

$$f(x) = -2(x - 3)^2 + 12$$

STATION 2 • TARGETED INSTRUCTION • 18 MIN

Standard Form: What Each Term Tells You

$$f(x) = ax^2 + bx + c, \quad a \neq 0 \quad f(0) = c \Rightarrow \text{y-intercept } (0, c) \quad \text{axis: } x = -\frac{b}{2a}$$

EXAMPLE A

$f(x) = x^2 - 6x + 5$:
name a , b , c and the axis

$$a = 1, \quad b = -6, \quad c = 5$$

y-intercept $(0, 5)$

$$x = -\frac{-6}{2(1)} = 3$$

EXAMPLE B

$f(x) = -2x^2 + 8x - 3$:
axis and vertex

$$x = -\frac{8}{2(-2)} = 2$$

$$f(2) = -8 + 16 - 3 = 5$$

vertex $(2, 5)$, max value 5

STATION 2 • TARGETED INSTRUCTION • 18 MIN

Graphing from Standard Form

vertex $\left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right)$ $(0, c)$ and its mirror $\left(-\frac{b}{a}, c\right)$ have equal height

EXAMPLE A

Graph $f(x) = x^2 - 4x + 3$

$x = 2$, vertex $(2, -1)$

y-intercept $(0, 3)$, mirror $(4, 3)$

$a = 1 > 0$: opens up; plot and join

EXAMPLE B

Graph $f(x) = -x^2 + 4x + 1$

$x = 2$, vertex $(2, 5)$

y-intercept $(0, 1)$, mirror $(4, 1)$

$a = -1 < 0$: opens down

EXAMPLE C

Same parabola:
 $(x - 3)^2 - 4$ in standard form

$$\begin{aligned}(x - 3)^2 - 4 &= x^2 - 6x + 9 - 4 \\ &= x^2 - 6x + 5\end{aligned}$$

STATION 3 • PRACTICE • 12 MIN

Give it a go: Practice, Apply, Investigate

PRACTICE

Secure the method

$f(x) = 3(x - 5)^2 - 2$: vertex, axis, direction

Vertex $(2, -3)$, through $(0, 1)$: write $f(x)$

Done when: the vertex is an ordered pair with the sign of h checked, and a is found by substitution and checked with the given point.

APPLY

Use it in context

Vertex $(3, -1)$, through $(5, 7)$: write $f(x)$

$g(x) = 2x^2 - 8x + 5$:
 a, b, c , axis, vertex, y-intercept and its mirror

Done when: every answer is checked by substituting a point, and g is sketched symmetric about its axis.

INVESTIGATE

Find out why

Expand $a(x - h)^2 + k$ and show $h = -\frac{b}{2a}$

$y = (x + 3)^2 - 4$:
why is the vertex not $(3, -4)$?

Done when: your argument uses expanding or the graph, not just a computed answer.

STATION 4 • PRODUCTION • 12 MIN

Production: Designing a Footbridge Arch

THE MODEL

A pedestrian footbridge in a Red Sea coastal tourism development, in the spirit of Vision 2030, has a parabolic arch. Standing on the left foot of the arch, the walkway rises to a highest point of 12 metres above the water at a horizontal distance of 20 metres, and the right foot is level with the left. Take x as the horizontal distance in metres from the left foot and y as the height in metres. These are invented model values for the classroom.

Vertex $(20, 12)$, left foot $(0, 0)$

$$y = a(x - h)^2 + k$$

1. Write the height y in vertex form, using the vertex and the left foot.
2. Expand to standard form, name a , b and c , and say what $c = 0$ means for the arch.
3. Use the axis formula to confirm the axis of symmetry, then find the height 5 metres from the left foot and the other point at that height.

DONE WHEN

- The vertex form is checked by substituting the left foot.
- The standard form is shown as an expansion with a , b and c named.
- The axis $x = 20$ is confirmed with the formula.
- Every answer has units (metres) and a sentence in context.

Hint for solving: put $(20, 12)$ in first, then use $(0, 0)$ to find a ; the right foot is the mirror image of the left foot.

STATION 5 • MASTERY GATE • 6 MIN

Mastery Gate: three questions to pass

1

Vertex $(-1, 4)$, through $(1, -4)$:
write $f(x)$

2

$f(x) = x^2 + 6x + 5$:
state c , the axis and the vertex

3

$f(x) = -x^2 + 2x + 3$:
direction, axis, vertex, mirror of the y-intercept

Pass mark

3 of 3 correct unlocks Station 6. Fewer than that: return to the slide for the question you missed, redo it, then re-take the gate.

STATION 6 • SMART PRODUCTION • 6 MIN

Smart Production: make one, then check it

CREATE

1. Invent a parabola in the first quadrant: choose a vertex and one extra point, and write it in vertex form.
2. Expand it to standard form and list a , b , c , the axis, the vertex and the y -intercept.
3. Swap with a partner and check that both forms agree by substituting a point.

Done when: both forms are written, the partner's check is signed, and one sentence says which form was quicker for finding the vertex.

TIME TO CHECK

Exit ticket: write your answers on a sticky note and place it on the door.

$$f(x) = x^2 - 8x + 10 : \text{axis and vertex}$$

Vertex $(4, -6)$, $a = 1$: write $f(x)$

Next lesson: 2-3 Factored Form of a Quadratic Function.

Exam connections

SAT

$h(x) = 2x^2 - 12x + 7$:
 x -coordinate of the vertex?

A) -3 B) 2 C) 3 D) 6

STEPS

$$1 \quad x = -\frac{b}{2a} = -\frac{-12}{2(2)}$$

$$2 \quad x = \frac{12}{4} = 3 \Rightarrow \text{C)}$$

GAT

$A =$ least value of $x^2 - 6x + 10$
 $B = 1$

STEPS

$$1 \quad x^2 - 6x + 10 = (x - 3)^2 + 1$$

$$2 \quad (x - 3)^2 \geq 0 \Rightarrow \text{least value } 1$$

$$3 \quad A = B$$

SAAT

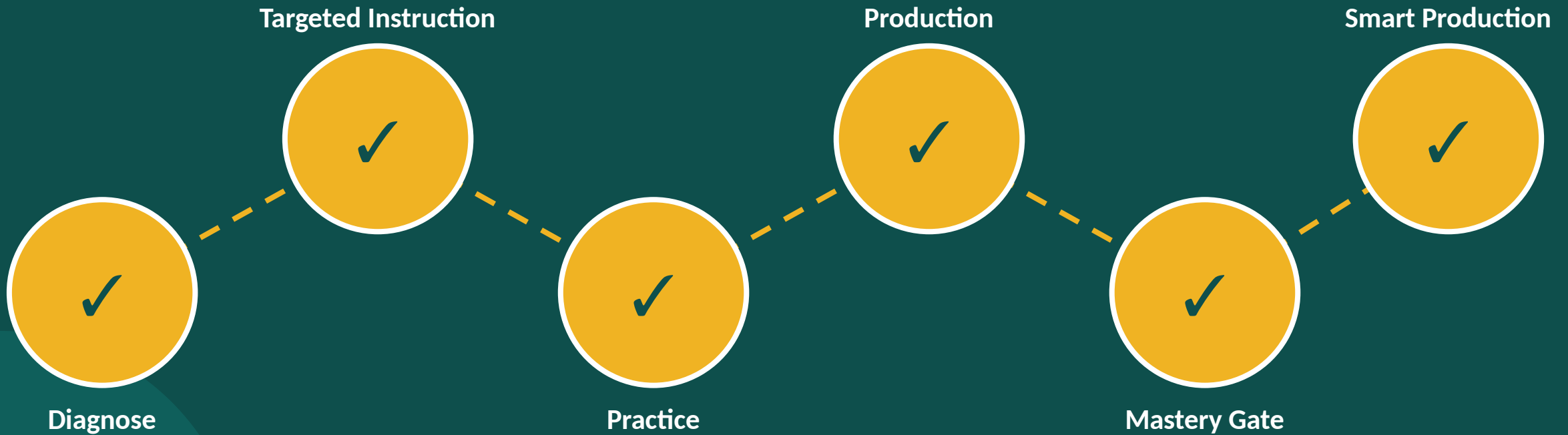
Maximum value
of $f(x) = -x^2 + 6x - 5$
A) 3 B) 4 C) -5 D) 6

STEPS

$$1 \quad x = -\frac{6}{2(-1)} = 3$$

$$2 \quad f(3) = -9 + 18 - 5 = 4 \Rightarrow \text{B)}$$

Journey complete



A parabola has two views: vertex form shows its turning point, standard form shows its y-intercept, and the axis $x = -b/(2a)$ links them.